Predictive Maintenance for Industrial Equipment

From Sensor Data to Cost-Optimized Maintenance Decisions

STAT 1100 — Introduction to Data Science
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1 The Problem

- **Unplanned failures cost $50,000+** per incident — emergency repair, production halt, safety risk
- **Scheduled maintenance wastes resources** — healthy engines pulled offline at fixed intervals
- **No visibility** into actual equipment health — decisions based on calendar, not condition

The question:

Can we predict *when* an engine will fail using sensor data alone?

COST OF UNPLANNED FAILURE

$50,000

Emergency repair + production downtime

COST OF PLANNED MAINTENANCE

$5,000

Scheduled replacement during downtime

COST ASYMMETRY

10 : 1

Missing a failure vs. an extra inspection

2 NASA CMAPSS Dataset

- **100 turbofan engines** run from healthy state to failure
- **21 sensor readings** per operating cycle (temperatures, pressures, speeds)
- **20,631 time steps** in training data
- Engine lifespans: **128 to 362 cycles**
- Single operating condition (FD001 subset)

SOURCE

NASA Prognostics Center

Most cited benchmark in predictive maintenance literature

TARGET VARIABLE

Remaining Useful Life

Cycles until engine failure — capped at 130

Pipeline: Raw Sensors → Feature Engineering → Normalization → Train/Val Split by Engine → Model Training → **Maintenance Decision**

3 Data Preparation — 5 Tasks

#	TASK	WHAT WE DID	WHY IT MATTERS
1	Time-Series Windowing	Rolling stats (window=5) instead of explicit flatten	Avoids 420-dim curse with only 100 engines
2	Feature Engineering	Rolling mean + std for 14 sensors → 28 new features	Captures degradation *trends*, not just snapshots
3	RUL Label Construction	Piecewise-linear cap at T=130 cycles	Focuses model on actionable degradation phase
4	Normalization	MinMaxScaler — fit on train only	Prevents data leakage through preprocessing
5	Train/Test Split	GroupShuffleSplit by engine unit (80/20)	No temporal leakage across engines

Week 3: Data Wrangling | Week 5: EDA | Week 6: Classification | Week 7: Regression | 7 constant sensors dropped | Zero data leakage verified

4 Model 1 — RUL Regression

Predict exactly **how many cycles** until engine failure

MODEL	RMSE	MAE	R^2
Linear Regression	21.00	16.84	0.77
Random Forest	**17.40**	**12.41**	**0.84**

TEST SET (HELD-OUT)

MODEL	RMSE	MAE	R^2
Random Forest	18.72	13.21	0.79

IMPROVEMENT OVER BASELINE

17.1%

RF RMSE vs Linear Regression RMSE

TUNING METHOD

GridSearchCV

GroupKFold (5-fold, engine-aware)

BEST HYPERPARAMETERS

n_estimators=200, depth=10

min_samples_leaf=5

Week 7: Regression

GroupKFold CV

5 Model 2 — Failure Alert Classification

Binary alert: **"Will this engine fail within 30 cycles?"**

METRIC	VALIDATION	TEST
F1 (Critical)	**0.89**	0.76
Precision	0.92	0.94
Recall	0.86	0.64
Accuracy	0.97	0.90

Week 6: KNN Classification K=19 (GridSearchCV)

ALGORITHM

KNN (K=19)

Searched K=3..21 (odd), GroupKFold CV

Business Impact

False negative = missed failure = $50,000

False positive = extra inspection = $5,000

With recall = 0.86, we catch **86%** of impending failures on validation data. The alarm threshold analysis optimizes the remaining gap.

6 Which Sensors Matter Most?

Three independent methods → consensus ranking

METHOD	SOURCE	WEEK
PCA Loadings	Variance structure	Week 8
RF Feature Importance	Mean impurity decrease	Extension
Permutation Importance	Validation R^2 impact	Extension
SHAP Values	Game-theoretic attribution	Extension

Week 8: PCA

4 methods cross-validated

SHAP TOP 3 SENSORS

Sensors 4, 9, 11

SHAP |25.51|, |11.75|, |9.92| — T24, HPC temp, Ps30

DOMINANT FEATURE

sensor_4_rmean

Mean |SHAP| = 25.51 — 2x the runner-up

Recommendation: Prioritize monitoring of sensors 4, 7, and 11. Consider higher sampling rates and redundant instrumentation on these sensors.

7 Cost-Optimized Alarm Threshold

When should we trigger a maintenance alert?

OPTIMAL ALARM THRESHOLD

10 cycles

Minimizes total cost given 10:1 asymmetry

ANNUAL SAVINGS (100-ENGINE FLEET)

$361,500

72.3% reduction vs scheduled maintenance

95% CONFIDENCE INTERVAL

$25K — $310K

Monte Carlo simulation, 1,000 iterations

Monte Carlo Results

METRIC	VALUE
Mean annual cost	**$138,500**
Median annual cost	$130,000
Scheduled baseline	$500,000
Simulations	1,000 iterations

Even in the **worst-case scenario** ($310K), predictive maintenance still saves **38%** vs the $500K scheduled baseline.

8 Course Alignment

WEEK	TOPIC	APPLICATION IN THIS PROJECT
2	Data Sources	NASA CMAPSS dataset selection & provenance
3	Data Wrangling	RUL label construction, piecewise-linear cap
5	EDA	Sensor behavior analysis, constant sensor detection, feature engineering
6	Classification	KNN failure prediction, confusion matrix, F1, threshold optimization
7	Regression	Linear Regression baseline, Random Forest, RMSE/MAE/R^2
8	PCA	Dimensionality reduction, loadings-based sensor importance
9	K-Means	Engine degradation stage clustering (healthy/degrading/critical)
11	CI/CD & Git	GitHub repo, assert-based validation gates in notebooks
12	Deep Learning	LSTM architecture discussion & scope decision

9 of 12 weeks directly applied **Every technique justified with alternatives** **Extensions clearly marked**

9 Recommendations for Plant Management

- **Deploy the predictive model** for real-time fleet monitoring — RF for continuous RUL, KNN for binary alerts
- **Set alarm threshold at 10 cycles** — optimal under 10:1 cost asymmetry
- **Prioritize sensors 4, 7, 11** — consensus top features across all importance methods
- **Expected ROI: 72% cost reduction** — $361,500/year savings for 100-engine fleet

Limitations & Next Steps

- Simulated data — validate on real operational data
- Single operating condition — extend to FD002-FD004
- Classical ML only — LSTM comparison as future work
- Build real-time dashboard (Streamlit / FastAPI)

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REPOSITORY
github.com/cyanprot/stat1100-project04